

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1267

Country Club 3 100 kV
October 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1267. The Interconnection Customer proposes a 173.0 MW solar photovoltaic facility interconnecting at the Country Club 3 100 kV substation in the Indiana area. The request was received on August 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 450 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,096,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BLUFFTON / RANDVILLE 100 KV
Contingency: SIGNETICS SUBSTATION / COUNTRY CLUB 3 100 KV L
Loading: 373.8% (Rating: 101 MVA)
DFAX: 100.0%
Pre-project loading: 203.1%
2. Facility: BLUFFTON / AZLE 100 KV
Contingency: COUNTRY CLUB 3 / SIGNETICS 100 KV
Loading: 337.9% (Rating: 117 MVA)
DFAX: 100.0%
Pre-project loading: 189.6%
3. Facility: RANDVILLE / HOFMAN 100 KV
Contingency: SIGNETICS / COUNTRY CLUB 3 100 KV L
Loading: 318.8% (Rating: 113 MVA)
DFAX: 100.0%
Pre-project loading: 166.1%
4. Facility: SIGNETICS / COUNTRY CLUB 3 100 KV
Contingency: AZLE / COUNTRY CLUB 3 100 KV
Loading: 267.0% (Rating: 154 MVA)
DFAX: 100.0%
Pre-project loading: 154.9%
5. Facility: AZLE / COUNTRY CLUB 3 100 KV line
Contingency: COUNTRY CLUB 3 / SIGNETICS 100 KV
Loading: 267.0% (Rating: 154 MVA)
DFAX: 100.0%
Pre-project loading: 154.9%

6. Facility: SIGNETICS / MONTPELIER ELECTRIC GENERATING STA 100 KV line
Contingency: WABASH 500 kV/230 kV Transformer Bank
Loading: 228.0% (Rating: 142 MVA)
DFAX: 75.4%
Pre-project loading: 136.1%
7. Facility: SIGNETICS / HARTFORD CITY 100 KV L
Contingency: MONTPELIER ELECTRIC GENERATING STA / SIGNETICS 100 KV
Loading: 213.2% (Rating: 126 MVA)
DFAX: 72.3%
Pre-project loading: 113.8%
8. Facility: HARTFORD CITY SUBSTATION / HOFMAN 100 KV
Contingency: SIGNETICS / COUNTRY CLUB 3 100 KV line
Loading: 159.3% (Rating: 115 MVA)
DFAX: 54.6%
Pre-project loading: 77.4%
9. Facility: INTALCO / MONTPELIER ELECTRIC GENERATING STA 100 KV
Contingency: SIGNETICS / MONTPELIER ELECTRIC GENERATING STATION 100 KV line
Loading: 156.9% (Rating: 130 MVA)
DFAX: 59.7%
Pre-project loading: 77.4%
10. Facility: HOFMAN / INTALCO 100 KV
Contingency: SIGNETICS / COUNTRY CLUB 3 100 KV
Loading: 134.1% (Rating: 119 MVA)
DFAX: 45.4%
Pre-project loading: 68.3%
11. Facility: COSO 7 / W ASHEVILLE 100 KV
Contingency: MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500 KV L
Loading: 125.5% (Rating: 178 MVA)
DFAX: 14.7%
Pre-project loading: 111.2%
12. Facility: COSO 7 / MONTPELIER ELECTRIC GENERATING STA 100 KV L
Contingency: MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500 KV line
Loading: 107.1% (Rating: 230 MVA)
DFAX: 14.7%
Pre-project loading: 96.0%
13. Facility: HARTFORD CITY / BEDFORD RURAL 100 KV
Contingency: SIGNETICS / MONTPELIER ELECTRIC GENERATING STATION 100 KV Line
Loading: 102.0% (Rating: 116 MVA)
DFAX: 40.8%
Pre-project loading: 41.2%
14. Facility: HILLTOP 500-345 kV Transformer Bank
Contingency: TALENENERGY MARTINS CREEK LLC WILLIAMSPORT / HILLTOP 500 KV
Loading: 100.6% (Rating: 2,218 MVA)
DFAX: 6.6%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUFFTON / RANDVILLE 100 KV	Rebuild line with higher-capacity conductor	\$4,942,000	54% (\$2,687,000)
BLUFFTON / AZLE 100 KV	Rebuild line with higher-capacity conductor	\$2,511,000	54% (\$1,365,000)
RANDVILLE / HOFMAN 100 KV	Rebuild line with higher-capacity conductor	\$18,286,000	54% (\$9,942,000)
SIGNETICS / COUNTRY CLUB 3 100 KV Line	Rebuild line with higher-capacity conductor	\$3,456,000	54% (\$1,879,000)
AZLE / COUNTRY CLUB 3 100 KV line	Rebuild line with higher-capacity conductor	\$12,624,000	54% (\$6,864,000)
SIGNETICS / MONTPELIER ELECTRIC GENERATING STA 100 KV line	Rebuild line with higher-capacity conductor	\$2,650,000	48% (\$1,281,000)
SIGNETICS / HARTFORD CITY 100 KV line	Rebuild line with higher-capacity conductor	\$10,769,000	54% (\$5,855,000)
HARTFORD CITY / HOFMAN 100 KV Line	Rebuild line with higher-capacity conductor	\$4,657,000	54% (\$2,532,000)
INTALCO / MONTPELIER ELECTRIC GENERATING STA 100 KV	Rebuild line with higher-capacity conductor	\$8,682,000	53% (\$4,565,000)
INTALCO / HOFMAN 100 KV	Rebuild line with higher-capacity conductor	\$5,594,000	54% (\$3,041,000)
COSO 7 / WEST ASHEVILLE 100 KV	Rebuild line with higher-capacity conductor	\$5,303,000	33% (\$1,736,000)
COSO 7 / MONTPELIER ELECTRIC GENERATING STA 100 KV L	Reconductor line	\$2,688,000	33% (\$880,000)
HARTFORD CITY / BEDFORD RURAL 100 KV	Reconductor line	\$2,199,000	53% (\$1,158,000)
HILLTOP 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$26,915,000	1% (\$311,000)
	Total allocated to project		\$44,096,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.